CSL4050

Computer Graphics

Introduction to the Course

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Reference Slides of Dr. Avinash Sharma

Start Smart: Why, What, How

Why Study Computer Graphics? Why it matters?

What is Computer Graphics Fundamentally?

How it works?

1. Why Study Computer Graphics?

Is this Real or Computer Generated?

- **Movies and TV:** From subtle digital makeup and set extensions to fully computer-generated characters and worlds (e.g., *Avatar*, Marvel Cinematic Universe).
- **Video Games:** The driving force behind real-time graphics innovation. Modern games feature incredibly complex and realistic environments and characters, all rendered in real-time.

Is this Real or Computer Generated?

- **Computer-Aided Design (CAD) / Computer-Aided Manufacturing (CAM):** Essential for designing and manufacturing everything from consumer products to airplanes.
- **Architectural and Interior Design:** Creating photorealistic visualizations of buildings before they are built, allowing for virtual walkthroughs and design validation.

This computer-generated image, created by Enrico Cerica using OctaneRender, shows ray traced glass distortion in the light fixture, diffuse lighting in the window and frosted glass in the lantern on the floor reflection on the frame picture.

Augmented Reality

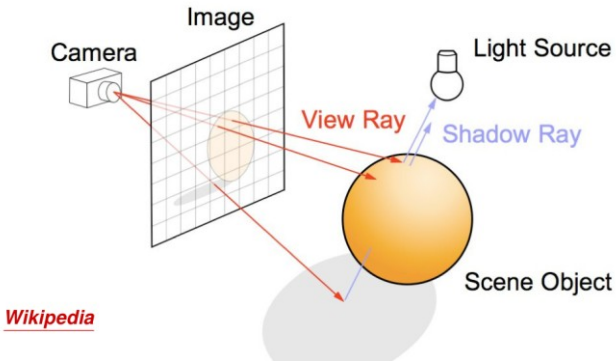
Metaverse

Meta Inc.

2. What is Computer Graphics?

Techniques and tools to generate realistic images using computers

Image Formation



- Represent physical or imaginary worlds computationally
- Apply mathematical models to simulate light and appearance
- Render images fast enough for real-time interaction
- Bridge between human perception and digital computation

3. How to achieve realistic images?

Create representations and "models" of the world

Create algorithms to produce ultra-realistic images

Do these fast

Producing Realistic Images

1. Represent physical world using basic primitives

- Basic geometric shapes and objects
- Efficiency vs utility: Use simple and useful primitives
- Break up complex scenes into available components

2. Transform a primitive to a camera image correctly

- Apply geometrical pin-hole camera model on the primitive
- A series of computations to map primitive to image
- Paint the picture based on physical properties of objects

3. Apply the same to the scene consisting of primitives

- Evaluate how multiple primitives interact or interfere
- Paint the physically correct picture of the whole scene

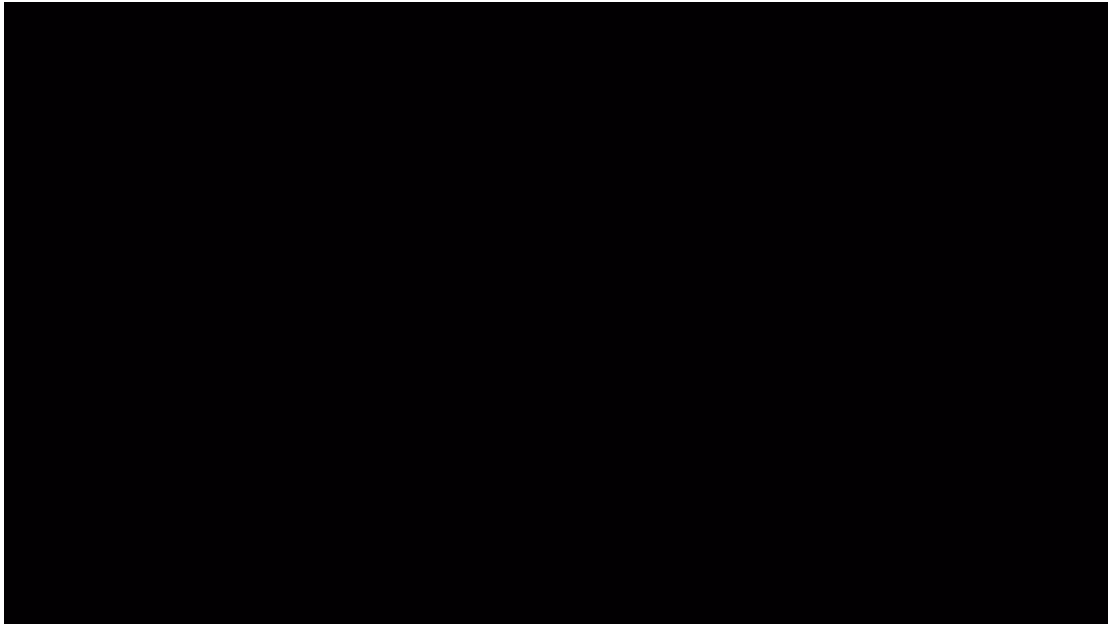
Continued...

Producing Realistic Images

4. Do all this efficiently

- Millions of primitives. High resolution images
- Complex objects with fine structure and properties
- Update image fast for application
- Real-time for games

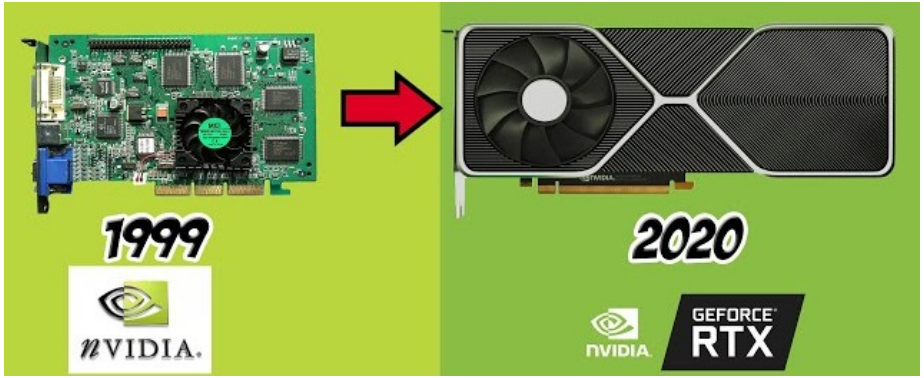
Game Animation Sample



Epic Games

GPU Revolution

- Graphics became "popular" only after mass-produced Graphics Processing Units (GPUs) or graphics accelerators came into existence
- **Graphics Accelerators**: On board hardware to speed up graphics computations
- Accelerators were expensive until end nineties!

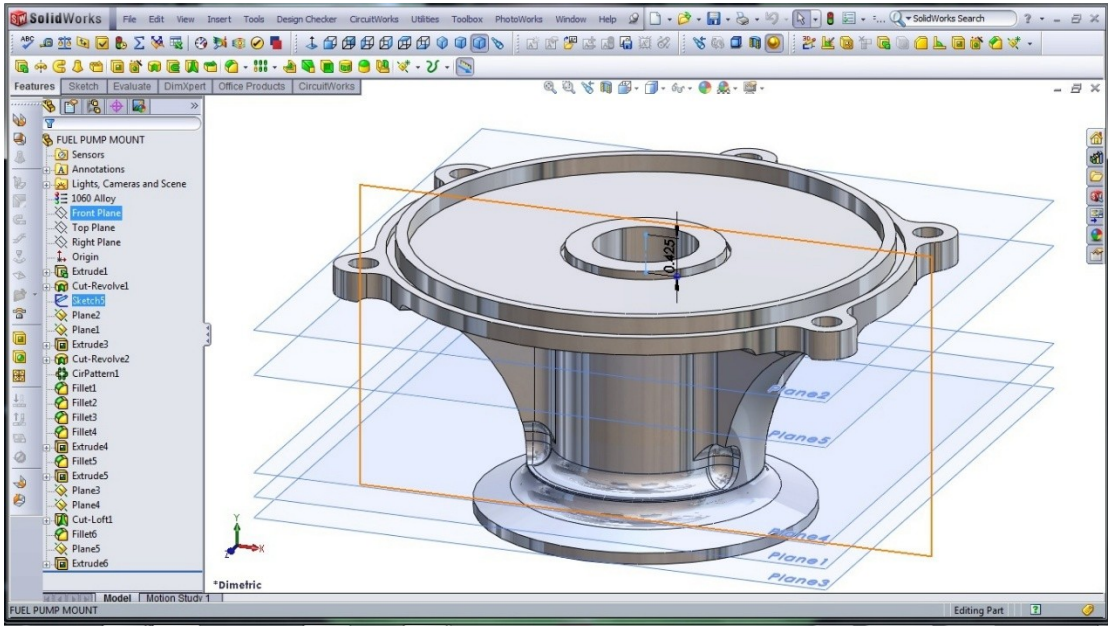


Computer Graphics Areas

Modelling, Rendering and Animation

Modelling

Deals with the mathematical specification of shape and appearance properties in a way that can be stored on the computer



Rendering

Deals with the creation of shaded images from 3D computer models

Animation

Engineer Explain in hindi @EngineerExplaininhindi

- A technique to create an illusion of motion through sequences of images
- Animation uses modelling and rendering but adds the key issue of movement over time, which is not usually dealt with in basic modelling and rendering

Graphics Programming

- **OpenGL**: current ANSI standard
  - Evolved from SGI's GL (graphics library)
  - Window system independent programming
  - GLUT (utility toolkit) for the rest
  - Popular: Many accelerators support it
- **DirectDraw/Direct3D**: Microsoft's attempt at it!
- **WebGL**: OpenGL to be used for web programming that is now gaining popularity
- **OpenGL ES**: Slightly reduced version for mobile devices, which will be the prime computing platform
- **Vulkan**: Emerging universal platform for Graphics

Course Logistics

Structure, Content, Books & other important information

Course Structure & Credits

- Structure (L-T-P [C]): 3-0-0 [3]
  - 3 Lecture Hours/Week
  - Slot H – LHC 106
- Assessment:
  - Project (10%): Group
  - Lab Assignment (20%): Weekly labs with deliverables
  - Quiz: 10%, twice
  - Midterm Exam: 20%
  - Endterm Exam: 40%

- Enrol on Google Classroom
  - Class Code: jqn3enpl

Course Content

Introduction

- 2D: Graphics Pipeline & Primitives
- 3D: Graphics Pipeline & Primitives
- Transformations: Projection & Viewing

Lighting & Shading

- Clipping and Rasterization
- Visibility
- Culling
- Buffering
- Binary Space Partitioning

Ray Tracing & Beyond

- Texture Mapping
- Shadows
- Ray Tracing and Global Illumination
- 3D Modelling
- Curve Fitting
- Introduction to Animation

Tutorial Content

APIs

- Introduction to Graphics APIs: OpenGL and Vulkan
- OpenGL: Building 2D World
- OpenGL: Building 3D World

Game Engine

- Vulkan: A Hands-on Tutorial
- Introduction to Unity
- Building a Simple Game in Unreal Engine

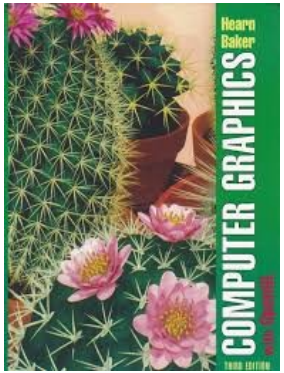
Advanced Topics

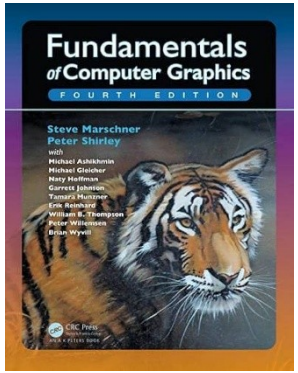
- Introduction to Blender
- Advanced Rendering Topics and Future Trends

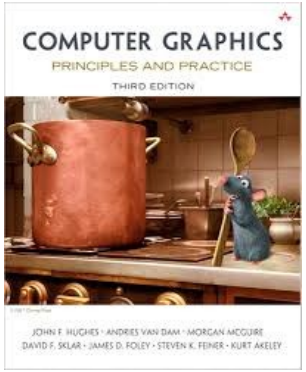
Required Background

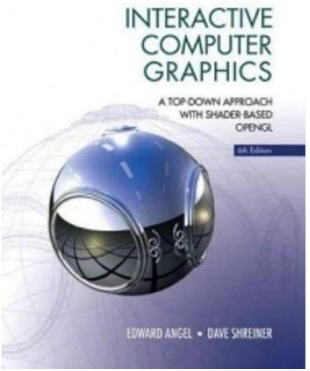
- Good programming skills in C/C++, Python
- Trigonometry basics
- Data structures
- Geometry/Linear Algebra: Points, vectors, matrices, transformations, etc
- Good imagination: Ability to visualize in 3D

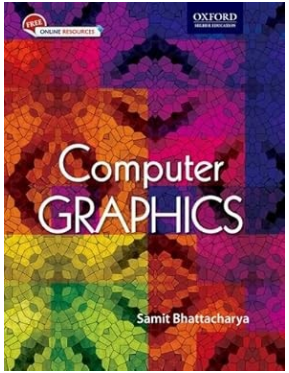
Learning Materials

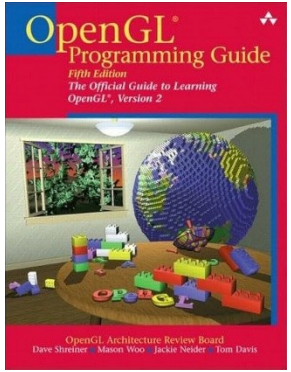












Important Information

Mode of delivery

- PowerPoint Slides
- Board Drawings
- Mandatory Interaction

Class Don’ts

- Do not use mobile phones inside class
- Do not come late
- Do not disturb the class with un-necessary chit-chat
- Do not play games in the class
- Do not copy from your friends

Thank You

Let’s build something amazing